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A Hybrid Intrusion Detection System using Machine Learning

Dr. K. Sangeetha[1], Dr. P. Vishnu Raja[2], Pradeep T[3], Sabari M[4] and Sarathi P[5]
[1,3-5]Kongu Engineering College/ Department of CSE, Erode, India
Email: sangeetha_k.cse@kongu.edu, pradeept.21cse@kongu.edu, sabarim.21cse@kongu.edu, sarathip.21cse@kongu.edu
[2]Builders Engineering College/ Department of AI&DS, Erode, India
Email: vishnurajap@gmail.com

***Abstract*—Because of the expanding volume of traffic and the increasing sophistication of cyberattacks, network security is becoming an increasingly difficult task. We solve this by introducing a hybrid intrusion detection system that efficiently detects network anomalies by utilising feature selection approaches and cutting-edge machine learning techniques. Recursive feature elimination is used in the suggested approach to repeatedly eliminate less important features, concentrating on the most significant features and improving model performance. To further optimise the feature set and reduce noise, SelectKBest is also utilized to find features with the most statistical importance. Modern machine learning models like Random Forest, KNN, and XGBoost are used for anomaly detection; an ensemble classifier combines their distinct advantages for increased robustness and accuracy. The effectiveness of the recommended strategy is demonstrated by experimental results utilising the NSL-KDD dataset, which contains a range of network attack types and typical traffic patterns. The efficacy of the model was assessed using performance indicators like accuracy, precision, recall, and F-measure, highlighting the method's resilience. The hybrid approach is positioned as a significant development for bolstering network defences across industries due to its capacity to improve real-time detection of network anomalies and its potential for incorporation into larger cybersecurity frameworks.**

***Index Terms*—hybrid features selection, machine learning, ensemble learning, anomaly detection, intrusion detection system, network security.**

I. INTRODUCTION

In an era where cyber dangers are on the rise, anomaly detection in network traffic is essential for preserving the security and integrity of information systems. Because network behaviour is dynamic, traditional security measures frequently fail to detect fresh assaults, requiring the use of advanced detection techniques. The Network Security Laboratory - Knowledge Discovery and Data Mining (NSL-KDD) dataset, which offers a wide range of features representing both legitimate and malicious network activity, acts as a standard for assessing anomaly detection methods. By using cutting-edge feature selection strategies and contemporary machine learning algorithms involving Random Forest, K-Nearest Neighbours (KNN), and XGBoost. This study seeks to improve the detection of network anomalies. To find the most important and pertinent features, feature selection methods like SelectKBest and Recursive Feature Elimination (RFE) are used. This lowers noise and enhances model performance.

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This study shows how machine learning models can improve finding accuracy and robustness against changing network threats by utilising these optimised feature sets.

II. RELATED WORKS

In intrusion detection systems, machine learning models offer unique advantages over traditional approaches, particularly through Decision Trees by Zahedi Azam [1].

These models improve detection with adaptable, interpretable, and efficient real-time capabilities, enhancing cybersecurity and developing robust detection frameworks. Using the Hunger Games Search and Remora Optimization Algorithm for IoT networks, R. Kumar [2] presents a hybrid intrusion detection model, demonstrating improvements over conventional techniques in tackling IoT security issues. The need for efficient Intrusion Detection Systems (IDS) is fueled by the serious cybersecurity concern of intrusion attacks. In their study, W. Alghamdi et al.

[3] suggest an anomaly-based intrusion detection system (IDS) that uses information gain-rank (IG-R) and machine learning to optimize detection procedures and expedite feature selection utilizing classifiers such as random forest. The complementing functions of 5G and Wi-Fi 6 in wireless connectivity are examined by Oughton et al. [4]. 5G is recommended for public cellular use, whereas Wi-Fi 6 is best suited for private networks. The study emphasizes the security issues for intelligent devices and the crucial role IDS plays in detecting internal assaults that compromise IoT nodes. The usefulness of ensemble learning for intrusion detection systems (IDS) is highlighted by Adhi Bayu Tama et al. [5].

They classify IDS approaches and provide the "Stack of Ensemble" (SoE) model, which combines many base classifiers to improve anomaly detection efficiency and accuracy, after examining 124 papers. According to Muhammad Erza Amina et al. [6], the quick expansion of IoT devices due to mobile technology raises security threats in public Wi-Fi networks, such as impersonation attacks. By lowering the dimensionality of the data and increasing the precision and effectiveness of threat detection in IoT networks, feature learning helps to reduce these risks. In order to overcome the problem of limited labelled data, Cheng, Xu, Zhong, and Liu [7] suggest the Hierarchical Stacking Temporal Convolutional Network (HS-TCN) for anomaly detection in IoT traffic. HS-TCN improves detection efficiency and accuracy by using both labelled and unlabelled data, automatically filtering records that are unclear.

According to experimental findings, HS-TCN enhances IoT anomaly detection performance and processing efficiency. Li et al. [8] propose a Network Intrusion Detection System (NIDS) based on machine learning that uses Automatic Feature Selection and Two-Phase Hybrid Ensemble Learning to tackle high-dimensional data and complex attacks. Their framework, tested on large datasets for both wired and wireless networks, outperforms similar methods in detection accuracy. An IDS framework that combines an ensemble of C4.5, Random Forest, and Forest PA classifiers with the CFS-BA method for dimensionality reduction is proposed by Zhou et al. [9]. On certain datasets, their CFS-BA-Ensemble approach performs better than current IDS methods. A stochastic gradient descent meta-classifier is combined with lightweight classifiers such as Naive Bayes, logistic regression, and decision trees in Kumar, Thakur, and Ayyagari's [10] ensemble learning-based intrusion detection system.

The model performs better than individual classifiers in both binary and multiclass classifications, as demonstrated by its evaluation on the KDD Cup 1999, UNSW-NB15, and CIC-IDS2017 datasets. For anomaly-based intrusion detection, Tama et al. [11] suggest a stacked ensemble model that makes use of reliable base learners such as random forest, gradient boosting, and XGBoost.

The model increases accuracy and decreases false positives, offering insightful information for improving web attack detection. To enhance attack detection, Sharma and Yadav [12] provide an intrusion detection system (IDS) that makes use of machine learning and Recursive Feature Elimination (RFE).

Their methodology outperforms current methods in terms of classification accuracy when tested on the KDD CUP99 dataset. It incorporates decision trees, SVM, and random forest classifiers. Zhou et al. [13] suggest an IDS architecture that uses an ensemble of classifiers (C4.5, RF, Forest PA) with a voting mechanism and the CFS-BA algorithm for feature selection.

The strategy enhances detection accuracy and adaptability to novel attacks, exceeding existing methods. In their review of machine learning's use in intrusion detection systems, Thangaraj et al. [14] compare intrusion classification algorithms such as LDA, CART, and Random Forest.

The paper examines IDS efficacy in big data, 5G networks, and IoT scenarios, showcasing enhanced security threat detection with machine learning algorithms. Six classifiers for identifying malicious activity in Mobile Ad hoc Networks (MANETs) are compared by Pastrana et al. [15].

III. PROPOSED WORK

A. Experimental Setup

In this study, we utilize several libraries, including NumPy, pandas, and scikit-learn, to facilitate our machine learning tasks. All models are trained within a consistent framework, ensuring uniformity in the experimental setup. The experiments are conducted on a computer equipped with an Intel Core i5 processor, providing sufficient computational power for model training and evaluation. Our primary focus is on conducting multi-class classification experiments, using the NSL-KDD dataset, which is a benchmark for network intrusion detection. The dataset is divided into two portions: 80% is allocated for training set, while the remaining 20% is used for testing set. Each instance in the dataset undergoes preprocessing steps, such as normalization and label encoding, to ensure that the models can effectively process the data and deliver comprehensive results. This setup allows for a thorough evaluation of the machine learning models' ability to detect normal and anomalous network traffic with high accuracy and efficiency.

B. Dataset Discussion

In this study, we preprocess the NSL-KDD dataset, which comprises 41 features representing various network traffic instances categorized into normal traffic and several types of cyberattacks. The focus is on classifying these instances into four primary categories: Denial of Service, Probe, Remote to Local, and User to Root, along with normal traffic. The dataset which undergoes further transformation to classify the different types of attacks into broader categories. Specifically, DoS attacks (neptune, smurf), Probe attacks (ipsweep, nmap), R2L attacks (ftp_write, guess_passwd), and U2R attacks (buffer_overflow, rootkit) are grouped into distinct classes. Each of these categories is assigned a numerical label for easier classification by machine learning models. In addition, as mentioned in Table I, normal network traffic is labelled separately to distinguish it from attack instances.

TABLE I. ATTACK CLASSIFICATION

Attack Type	No of Instances in Training Set	No of Instances in Testing Set
Normal	60000	15000
DoS	30000	7500
R2L	20000	5000
U2R	5000	1250
Probe	25000	6250

C. Classification

Multi-class classification is employed to classify network traffic into distinct categories: normal and various types of attacks. Each data point belongs to one and only one class, and the objective is to accurately assign the appropriate class label based on the feature attributes. The models aim to detect network anomalies, providing crucial insights for cybersecurity by classifying instances as normal or as specific types of attacks.

D. Feature Engineering

Feature engineering concentrates on enhancing the performance of the model by adding new features or changing current ones. In order to find prospective features that could offer deeper insights into the data, this approach requires subject expertise. Scaling (normalizing or standardizing feature values), use methods such as label encoding or one-hot encoding to encode categorical data and creating interaction features (combining two or more features to capture relationships) are some examples of feature engineering techniques. Since feature engineering helps computers better comprehend the links and foundational trends in the data, it can improves the accuracy of the model. In this proposed work, we incorporated the one hot encoding for the text-based feature to separate them based on their labels, and standard scalar for converting them to standardize the input data.

E. Hybrid Feature Selection

Hybrid Feature Selection finds the most pertinent features for efficient prediction by combining the benefits of filter and wrapper approaches. Filter techniques, which use statistical metrics such as chi-square tests or correlation coefficients, offer a preliminary screening to rank and eliminate redundant or unnecessary characteristics. Then come wrapper methods like Recursive Feature Elimination, which iteratively improve feature selection by using a machine learning model to assess subsets in order to optimize prediction accuracy. Given the high degree of dimension and volume of network traffic data, hybrid feature selection is essential in the context of NIDS. It lowers the computational overhead by removing noisy or unnecessary features, allowing for quicker model inference and training. It also improves the model's capacity to concentrate on the most selective characteristics, which results in higher detection rates for a variety of attack types, even those that are infrequent or imitate typical traffic patterns. Because of this all-encompassing strategy, which guarantees a balance between classification accuracy and computing economy, hybrid feature selection is especially advantageous when developing strong and dependable intrusion detection systems. Feature selection techniques incorporated in the system are given below:

Recursive Feature Elimination

Recursive Feature Elimination (RFE) is a powerful wrapper feature selection technique that enhances model performance by systematically eliminating the least significant features from a dataset. In the context of using Random Forest, RFE begins by training the model on the complete feature set. Following training of the first model, RFE assesses each feature's significance according to how much it adds to the prediction ability of the model. This evaluation is typically derived from the feature importance scores provided by the Random Forest algorithm, which indicates how much each feature contributes to reducing impurity in the decision trees. As illustrated in Figure 1, after evaluating the importance of the features, RFE progressively eliminates the least significant one and retrains the model using the remaining features. This iterative process of elimination and retraining continues until the optimal subset of features is determined. The result is a reduced feature set that retains only the most critical attributes for the task at hand. By concentrating on these essential features, RFE lowers computing cost and dimensionality while simultaneously improving the Random Forest model's accuracy.

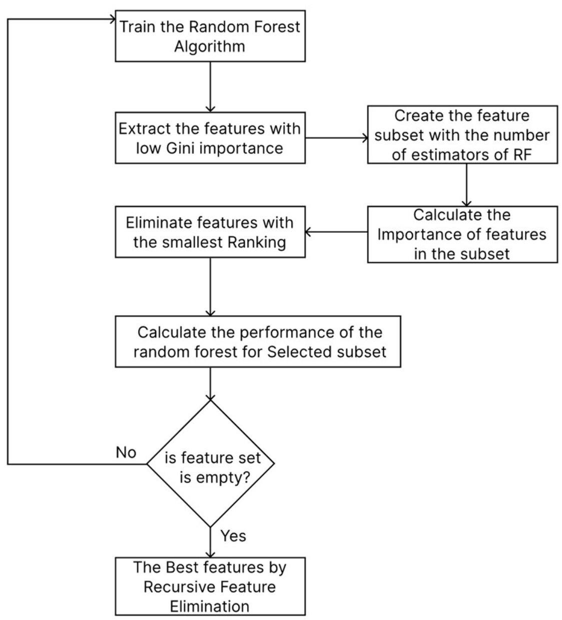


Figure 1. Workflow of the RFE-RF

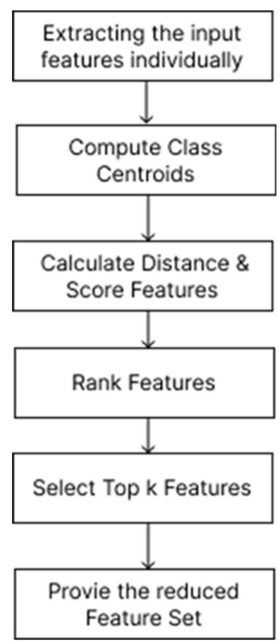


Figure 2. Workflow of the KBest

KBest Feature Selection

KBest feature selection is a widely used technique that adopts a filter method to assess the relevance of each feature independently of the classification model employed. This approach emphasizes identifying the most informative features by evaluating their statistical relationships with the target variable. KBest utilizes various scoring functions tailored to the nature of the features being analysed; for instance, Chi-squared tests are often applied for categorical features, while Analysis of Variance (ANOVA) is used for continuous variables. By quantifying how well each feature correlates with the predicted classes, KBest effectively determines which features play a significant role in distinguishing between different classes within the dataset. During the feature selection process, KBest ranks all features based on their calculated scores, selecting the top k features with the highest significance for inclusion in the model. This method allows for the retention of features that exhibit strong relationships with the target variable, thereby streamlining the dataset and reducing its dimensionality. By eliminating less informative features, KBest enhances the performance and effectiveness of the machine learning models employed, particularly in tasks such as anomaly detection. In addition to reducing training and inference durations, a more condensed and targeted feature set improves the model's capacity to generalize to previously unseen data. The Workflow of the KBest Algorithm is shown in Figure 2.

F. Training Models

After feature selection, KNN, Random Forest, and XGBoost models—each of which contributes unique strengths to the task of network intrusion detection—are trained using the ideal feature set. KNN is very useful for finding local patterns and anomalies in the data since it measures the distances between new occurrences and preexisting examples. KNN performs exceptionally well in situations where minute variations in value of features are crucial for classification because of its proximity-based methodology. In contrast, Random Forest uses a bagging strategy to reduce variation and improve model stability by combining predictions from many decision trees. It is ideally suited for complicated, multi-feature situations due to its capacity to assess feature relevance and manage high-dimensional datasets, guaranteeing accurate classification across a variety of network traffic patterns.

XGBoost, on the other hand, uses sophisticated gradient boosting algorithms to build decision trees sequentially, with each tree being constructed to fix the mistakes of the one before it. By concentrating on the most challenging cases to categorize, this iterative process helps XGBoost to attain high accuracy. It is especially useful for identifying complex patterns in the data because of its built-in regularization methods, which assist avoid overfitting. Additionally, XGBoost is a vital tool for identifying low-frequency but crucial attack types because to its effectiveness in managing large-scale datasets and its capacity to rectify class imbalances. When combined, these models leverage feature selection's advantages to offer a strong foundation for identifying and categorizing abnormalities in network traffic.

G. Ensemble Training

One effective machine learning method is ensemble learning, which integrates several separate models to provide a more reliable and accurate prediction system. The main objective of ensemble learning is to use the variety of different methods or variants of the same algorithm to enhance the overall effectiveness of models. This approach can mitigate the weaknesses of individual models, lowering the possibility of overfitting and enhancing the ability to generalize to new data. In the context of anomaly detection, ensemble learning proves particularly effective, as it allows for the integration of various models, such as KNN and Random Forest and XGBoost, each contributing unique strengths to the classification process.

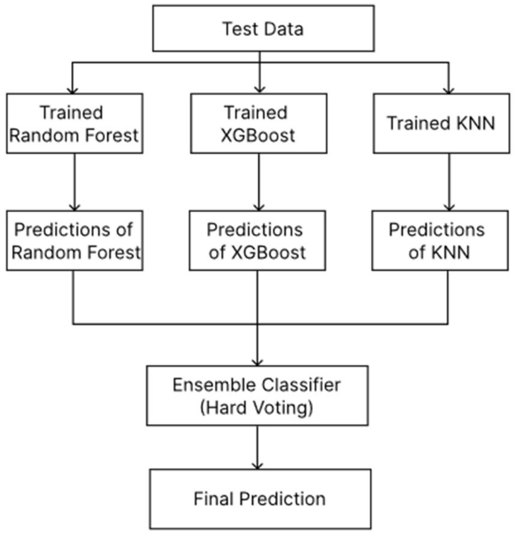
For instance, while KNN is quite good at identifying local patterns in the data, Random Forest is adept at handling high-dimensional feature spaces and evaluating feature importance and, XGBoost has the benefit of effective gradient boosting, which makes it possible to identify intricate patterns and reduce errors through repeated optimization.

These models work in concert to improve the anomaly detection system's overall accuracy and resilience. Ensemble learning improves the accuracy and dependability of classifications by utilizing strategies such as hard voting, in which the outcome of the prediction is determined by the combined models' majority vote. This synergistic approach enables the anomaly detection system to better adapt to the complexities and variations of network traffic, thereby increasing its effectiveness in identifying potential security threats. Additionally, ensemble learning can improve the robustness of the detection system by ensuring that even if one model fails to detect an anomaly providing a safety net for critical security applications. The workflow of Hard voting classifier for those models is shown in Figure 3.

H. Prediction

The prediction function is designed to randomly select four network traffic packets from the test dataset and use a trained machine learning model to predict whether each packet represents normal traffic or a type of network attack. The randomly selected packets are subjected to the model, which was trained on the labelled dataset. By contrasting the model's predictions with the real labels, their accuracy is assessed.

The selected packets are processed by the model, which outputs predictions in the form of numerical labels corresponding to the different traffic categories: Normal, DoS, Probe, R2L, or U2R.For each packet, the function prints its feature set along with the predicted attack type. Additionally, the true labels for the packets are also retrieved and mapped to the same attack categories, allowing for a direct comparison between the model's predictions and the actual attack type. This process aids in assessing how well the model classifies various kinds of network traffic, providing insight into how well the model generalizes to unseen data in detecting anomalies or attacks. The Complete workflow of the proposed work which includes the feature selection and ensemble model is shown in the Figure 4.



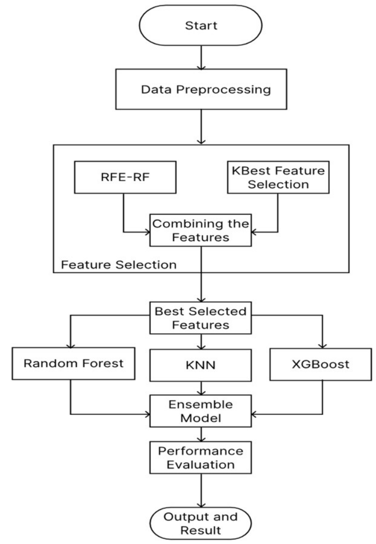


Figure 3. Voting Classifier (Hard Voting) Figure 4. Workflow of the Proposed Work

The proposed model for anomaly detection in network traffic integrates multiple phases, as depicted in the diagram. Data preparation, which includes operations like extracting features and transforming those features to get the dataset ready for analysis, is the first step in the process. Following preprocessing, the model applies a hybrid feature selection technique combining Recursive Feature Elimination (RFE) with Random Forest and KBest selection, allowing the identification of the most relevant features for classification. Once the feature set is optimized, the system utilizes ensemble learning, where each model that the KNN, XGBoost and Random Forest algorithms are employed. Each model is trained using the refined dataset, leveraging their respective strengths—KNN's local decision-making and Random Forest's robust ensemble structure—to accurately classify network traffic instances as normal or one of several attack types (DoS, Probe, R2L, U2R). The combined predictions from these models are then evaluated using various performance metrics, ensuring comprehensive detection of anomalies in network traffic.

IV. Parameters for Evaluation

A. Accuracy

In deep learning, accuracy is a commonly used evaluation parameter to determine a classifier's overall efficacy. It measures the percentage of both positive and negative predictions that were accurate out of all the guesses that were made. The ability of the model to classify images into various classes is demonstrated by accuracy, a complete metric for multi-class classification. It offers a comprehensive view of model's performance, particularly when handling intricate and varied classification problems.

$$\text{Accuracy} = \frac{(\text{TP} + \text{TN})}{(\text{TP} + \text{FN} + \text{FP} + \text{TN})} \quad (1)$$

Accuracy alone may lead to misinterpretation in certain situations, especially with imbalanced datasets or differing consequences of false positives and false negatives. In such cases, alternative metrics like recall and precision may provide more insightful analysis.

B. Precision

Precision in deep learning quantifies how effectively a classifier predicts positive examples. It is the ratio of true positive predictions to all predictions marked as positive by the classifier. Precision is important for minimizing false positives in classification tasks. It shows how well a model identifies positive cases accurately. Precision in multi-class classification is computed independently for each class and can be averaged using techniques such as macro- or micro-averaging. Precision measures the proportion of expected positive cases that are true. Accurate positive forecasts are shown by high precision, while low precision indicates many false positive predictions. Precision is crucial in tasks where false positives have serious consequences, such as healthcare diagnosis or fraud detection. A classifier with high precision ensures that positive predictions are accurate and reliable. Precision alone is insufficient to evaluate classifier performance as it doesn't account for missed true positives. Precision is often used alongside other metrics like recall and accuracy for a thorough assessment of classifier effectiveness.

$$\text{Precision(PPV)} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

C. Recall

Recall is a common evaluation metric. Out of all positive samples, it calculates the percentage of real positive cases that the model properly detected. In simpler terms, recall indicates how well the model can identify all pertinent positive cases. Recall is especially important when missing a positive case (true positive) is more costly than mistakenly identifying a negative case (false positive).
Mathematically, recall is calculated as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

D. F1-Score

When working with imbalanced datasets, the F1 score is a statistic used to assess a classification model's performance. It uses the proportional average of both recall and precision to balance the trade-off between the two. Recall gauges the capacity to record all pertinent positive occurrences, whereas precision measures how well positive predictions work.

$$\text{F1} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precison} + \text{Recall}} \quad (4)$$

V. RESULTS AND DISCUSSIONS

Each machine learning model undergoes training on the NSL-KDD training dataset, with validation carried out on a separate validation set. By predicting labels on the test data, the model's performance is subsequently assessed using the ideal model parameters that were saved during this process. In terms of training loss, validation accuracy, and test performance, each model demonstrates robust classification capabilities for network intrusion detection. When deploying these models on resource-constrained devices, such as edge devices, microcontrollers, or IoT devices, optimization of the model is crucial. Optimizing these models involves reducing their computational footprint while maintaining high accuracy. In the current study, various optimization approaches were explored. For instance, model pruning was applied to reduce the number of decision trees in Random Forest or remove less important features in KNN. Pruning reduces complexity and speeds up inference without significant loss in accuracy. Another technique, quantization, was employed, where model parameters such as decision thresholds were stored in lower precision formats during deployment. This technique significantly reduces the model size, making it more efficient for deployment on devices with limited memory or processing power, such as IoT sensors or mobile systems.

A. Comparison of Models on Each Feature Selection

Recursive feature elimination (RFE), that finds features according to how they affect a model's performance, is the first step in the assessment process. RFE chose 13 features by utilizing global trends in the data, and Random Forest achieved 99.801% accuracy. This approach works well for high-level feature selection since it is excellent at capturing more extensive relationships within the dataset. It might, however, miss small, regional patterns that are essential for identifying particular kinds of network irregularities. KBest, on the other hand, identifies 10 features that represent local patterns in the data by choosing features determined by statistical measurements like ANOVA F-values. As mentioned in the Table II, despite being computationally efficient, KBest's accuracy with Random Forest was lower at 96.679%. Its emphasis on localized relationships restricts its ability to translate, making it less effective in complicated contexts where feature interactions are critical. Proceeding to Principal Component Analysis (PCA), the method captured the highest variance in the dataset and converted it into 10 principal components. Models with PCA were 100% accurate, but interpretability suffered as a result. Since it becomes difficult to comprehend the underlying patterns, PCA is less useful for real-world applications due to its failure to preserve the original feature correlations. In certain situations, it could also result in overfitting, so it also missing some important information. Finally, the Hybrid Feature Selection (RFE + KBest) approach was applied, combining the global pattern recognition of RFE and the local pattern identification of KBest. This integration produced a comprehensive subset of 19 features, achieving the excellent results with an accuracy of 99.854% using Random Forest. By balancing global and local insights, the hybrid model demonstrated superior generalization and robustness in detecting both broad and intricate network intrusion patterns. The performance of the suggested ensemble model, which combines Random Forest, KNN and XGBoost to efficiently handle network intrusion detection, is highlighted the result analysis. The ensemble technique, which combines these models, demonstrates stability and reliability while excelling at spotting intricate infiltration patterns. The ensemble model showed better classification performance under different feature selection methods. Using Recursive Feature Elimination, for example, it outperformed individual models with an accuracy of 99.846%. Although PCA-based feature selection produced 100% accuracy, interpretability issues and possible overfitting limit its practical usefulness. The Hybrid feature selection method, which combines RFE and KBest, balanced feature efficiency and interpretability, achieving an ensemble accuracy of 99.854%, making it the most effective feature selection strategy. Figure 5 illustrates the comparative accuracy of all models across feature selection techniques, with the ensemble model consistently outperforming standalone approaches. These results validate the ensemble model's capability to generalize effectively across diverse network traffic scenarios, positioning it as a robust solution for real-world network intrusion detection systems.

TABLE I. PERFORMANCE ANALYSIS

Feature Selection	Random Forest	KNN	XGBoost	Ensemble Model
RFE	99.8	98.79	99.84	99.84
KBest	96.67	94.47	96.94	96.79
PCA	100	98.86	100	100
Hybrid	99.81	98.91	99.85	99.85

TABLE II. PERFORMANCE OF EACH ATTACK

Attack Types	Random Forest	KNN	XGBoost	Ensemble Model
DoS	99.93	99.43	99.98	99.76
Probe	99.38	94.21	99.63	99.38
R2L	95.15	94.85	96.06	96.97
U2R	31.25	37.50	62.50	56.25

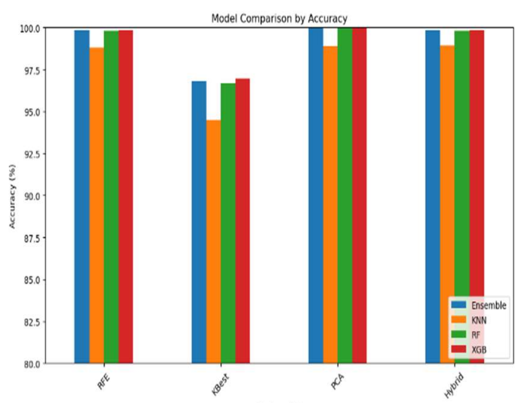


Figure 5. Accuracy Comparison of Each Model

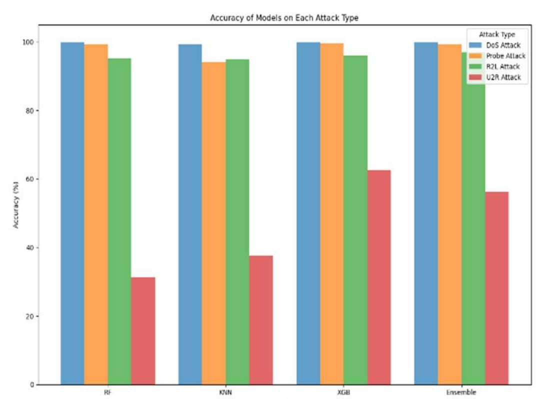


Figure 6. Accuracy comparison of all models with each attack

B. Comparision of each type of attack

The effectiveness of attack detection differs based on the used method and the type of attack. For instance, DoS attacks, with their high volume and easily identifiable traffic patterns, are generally well-detected by both signature-based and anomaly-based detection systems. Probe attacks, on the other hand, require more sophisticated models that can discern information-gathering activities hidden within normal traffic. R2L and U2R attacks are more challenging to detect due to their low frequency and their resemblance to normal user activities. Models that incorporate feature selection methods like Recursive Feature Elimination (RFE) and KBest tend to perform better at identifying these attacks by focusing on the most relevant and discriminative features. The Random Forest, KNN and XGBoost models, when combined in an ensemble, offer a more comprehensive solution, effectively handling different types of attacks by leveraging the strengths of each model in various scenarios. The usefulness of different models in identifying distinct attack types is demonstrated by the experimental findings, which are compiled in Table III and emphasize both their advantages and disadvantages. DoS assaults, which are distinguished by large traffic volumes and unique patterns, are most accurately identified by XGBoost (99.98%), closely followed by Random Forest (99.93%). The accuracy of the Ensemble model, which combines the best features of several classifiers, is 99.97%, while KNN comes in just behind at 99.43%. With respective accuracies of 99.63% and 99.63%, the XGBoost and the Ensemble model outperform Random Forest at 99.38% and KNN at 94.21% for probe assaults, which necessitate more sophisticated detection because of hidden trends in information-gathering operations. The Ensemble model has the highest accuracy of 96.97% in identifying R2L attacks, however they are more difficult to detect since they mimic normal user activity. XGBoost, Random Forest, and KNN follow with 96.06%, 95.15%, and 94.85%, respectively. With XGBoost attaining the best accuracy at 62.50%, the Ensemble model at 56.25%, and Random Forest and KNN trailing at 37.50%, U2R attacks are the rarest and most challenging to detect. While the Ensemble model shows its adaptability and balanced performance across all attack types, by combining the strength of each individual models to effectively handle the issues of intrusion detection systems, these findings demonstrate the resilience of XGBoost and Random Forest in high-accuracy tasks. The bar chart depicting accuracy of each attack against all the models is visualized in Figure 6.

VI. CONCLUSION AND FUTURE WORKS

Using the NSL-KDD dataset, this paper effectively demonstrates the design of a system for detecting anomalies for network traffic, highlighting the importance of feature selection, preprocessing, and combining multiple machine learning methods. By employing techniques such as Recursive Feature Elimination and KBest for feature selection, and by combining those feature selection as Hybrid method of feature selection, to achieve more feature with better accuracy, the study identifies critical attributes that enhance model performance. The potential for increased accuracy and dependability in identifying different network anomaly is demonstrated by the employment of Random Forest, XGBoost and KNN models in conjunction with an ensemble method. Ultimately, the suggested system not only advances the subject of network security but also provides a framework for further anomaly detection research, opening the door for the creation of more resilient and flexible security solutions that can change as cyber threats do. In future work, the anomaly detection system can be enhanced by incorporating advanced techniques such as deep learning algorithms, which may provide improved detection capabilities for more complex and evolving threats in network traffic. Additionally, exploring the integration of real-time data streams and implementing online learning methods would allow the system to adapt continuously to new patterns of behaviour, improving its responsiveness and accuracy. The strength and generalizability of the model would be improved by including varied network settings and broadening the dataset to contain a wider range of attack scenarios. Furthermore, investigating the application of hybrid models that combine the strengths of multiple algorithms could yield better performance metrics. Lastly, conducting user studies and implementing feedback mechanisms could refine the system based on practical experiences and evolving security challenges, ensuring its relevance and effectiveness in real-world applications. In future, we have to integrating threat intelligence feeds and contextual information, which can improve the system's ability to detect novel and zero-day attacks, in real-world applications.

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